

SAMANTHA FERNANDEZ-PEREZ

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EXPERIENCE

Research Intern, Dept. of Epidemiology

Weill Cornell Medicine, NY

May – Nov 2026

- Reproduced the SeNMo pan-cancer multi-omics survival pipeline across all 33 TCGA cohorts, preprocessing six modalities (clinical, mutation, RPPA protein, gene expression, DNA methylation, miRNA) to the authors' specification on the Cornell Cayuga HPC (Python, NumPy/pandas, Slurm).
- Built an independent 11,127-patient × 81,240-feature matrix — within 0.7% of the published 80,697 — and validated it against the authors' model: two of six omics matched exactly, with patient-level containment against the released cohort and a modality-by-modality feature cross-reference.
- Probed the model's 48-dimensional patient embeddings with cross-validated Cox models (concordance index) to evaluate how clinical and demographic signal is represented in the learned space.

Research Assistant, Dept. of Biochemistry & Molecular Biology

Syracuse, NY

Sep 2024 – Oct 2025

- Built a high-throughput screening system for prokaryotic peptide and protein libraries against SUMO and ULP therapeutic targets at Upstate Medical University SUNY.
- Optimized protein purification and molecular biology protocols, validating binding candidates to support scalable, cost-effective therapeutic development.

HB Experimental Forest Research Assistant

Woodstock, NH

Apr – Aug 2024

- Analyzed seasonal and elevational variations in particulate organic carbon and subsurface nanoparticles, documenting wollastonite's (CaSiO₃) role in mitigating acid rain.
- Led TACNY STEM Trekkers Summer Camp sessions on lake stratification and seasonal water chemistry.

Pediatrics Administrative Assistant

Queens, NY

Aug 2018 – Jun 2021

- Supported clinical staff with sterile room preparation and patient care logistics across pediatric appointments.
- Translated between English and Spanish for patient families, enabling clear communication during care decisions.

EDUCATION

Syracuse University, College of Engineering and Computer Science

Bachelor of Science in Biomedical Engineering

Coursework: Bioengineering Fundamentals*, Engineering Computational Tools*, Biomechanics, Medical Devices & Biomaterials, Quantitative Physiology, Biomedical Systems, Signal & Control*, Bioinstrumentation, Electrical Engineering*, Global Regulatory Affairs (* with Lab)

Extracurriculars: Syracuse University Waterpolo and Swim Club, Boxing Club, iGEM (International Genetically Engineered Machine), Women in Science and Engineering (WiSE), Society of Hispanic Professional Engineers (SHPE), Biomedical Engineering Society (BMES)

SKILLS

Software: Python, MATLAB, Jupyter, Anaconda, R-Studio, Fusion360, Canva, Microsoft Word, PowerPoint, Excel

Biolaboratory: Electroporation, DNA Gel Electrophoresis, Genetic Therapy, Heat Shock Transformation, Cell Culture

Environmental Lab: Ion Chromatography, Dynamic Light Scattering, Nitrogen and Carbon Analysis

Languages: Native Spanish, Fluent English, Conversational French and Portuguese

Certifications: CITI Health Information Privacy and Security for Clinical Investigators (HIPS)

ENGINEERING APPLICATIONS

Prosthetic Sixth Finger

Syracuse University

Jan – Feb 2024

- Designed in Fusion360 and programmed an Arduino-controlled prosthetic finger with temperature sensing and real-time patient feedback.

Coding Operating Python

Syracuse University

Jan – May 2023

- Built and tested a Python and microprocessor-based assistive device with two teammates, enabling computer control without fine motor input.

LEADERSHIP

iGEM President and CEO

Syracuse University

Jan 2023 – Present

- Founded Syracuse University's first iGEM team and recruited an interdisciplinary student group.
- Led proposal development for international competition across multiple academic departments.